

A Quick Scan

Also Remember...

- Upgrading your RAM is probably a good idea if you are going to be scanning high-resolution images. This will have a significant impact on overall scanning performance.
- Check for sRGB support. This is a new standard that will make Web images look even better.
- Consider buying a transparency adapter if your usage is high enough. Automatic document feeders (ADFs) are a boon to people with large text scanning needs.
- To get good results with 35-mm slides or negatives, you should look for a special scanner that has been designed for them.
- Check that your printer is good enough to reproduce the high quality scans that the scanner produces.

The scanner, which used to be a luxury not so long ago, is now well within the reach of most people. Prices have fallen partly due to competition between the numerous manufacturers and also due to the advancement of the technology used. This has put it high on the shopping lists for many home users, who basically use them to have fun, explore and maybe even compile their photo albums online. Today scanners begin at just about Rs 4,000 and offer pretty acceptable performance in terms of quality and speed. The presence of USB connectors on almost all new motherboards has helped ease the pain of installing the scanner at the same time as increasing scanning speed due to the higher data transfer rates it supports.

TECHNOLOGY INSIGHT

The main components in the scanner consist of a light sensitive diode, a light source, lens, the stepper motor, which moves the scan head and the mirror on which the document is placed. The light hits the document and the image is reflected back to the diodes via mirrors. The image is then focused on the CCD array, which converts the photon in to electron.

SMART TIPS

The light source: This is either a cold cathode fluorescent lamp (CCFL) or Xenon lamp. Xenon lamps give you better performance.

Resolution: The higher the resolution of the scanner, the sharper will be the image reproduced. Most low and mid-range scanners today have an optical (or hardware) resolution of 600x1200 pixels. Here the 600 refers to the number of CCDs or CIS (Contact Image Sensors) that are arranged in a horizontal row and 1200 infers that the stepper motor is capable of moving in steps of 1/1200th of an inch.



Platform size: For most home users, A4 size scanners are sufficient.

However, for professional uses a scanner with an A3 sized platform is preferable.

Colour depth: Though 36-bit is preferable, you should go for a minimum of 24-bit colour depth.

Connectors: If you have a free USB port in your system then a USB scanner is the recommended choice. Parallel port scanners have lower data transfer rates than the USB models, whereas the SCSI models, which require a special adapter, give you the best performance.

OCR software: Most bundled OCR software offer basic functionality. For heavy text-based scanning you should look at third-party OCR packages such as ABBY FineReader 5.0.

	Basic use	Intermediate use	Professional use
You need	Something that you will use occasionally use to scan home photos or something from a book or magazine	Something for more serious scanning, either for text documents or images	A scanner with high resolution and speeds because of large volumes of scanning (say, publishing, legal, graphics)
Look for	Any A4 scanner with USB support and a minimum resolution of 600x1200 dpi	Any A4 scanner with USB support and a minimum resolution of 1200x2400 dpi or better	A scanner with 1600x3200 dpi or better resolution, A3 platform size and preferably a SCSI interface
Our pick	Acer ScanPrisa 640U, UMAX Astra 2000p	HP ScanJet 5300C	Epson 1640SU
Price range	Rs 4,000 to Rs 10,000	Rs 10,000 to Rs 18,000	Above Rs 18,000



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